Plane Wall with Convection using One-Term Approximation

Saturday 10th October, 2026

Curvenote

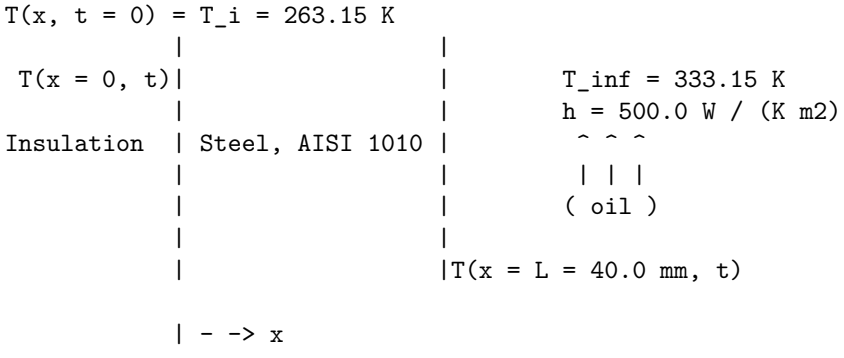
1 Problem Statement

2 Assumptions

- Pipe wall can be approximated as plane wall, since thickness is much less than diameter.
- Constant properties.
- Outer surface of pipe is adiabatic.

3 Schematic

- $\mathrm{k} = 63.9\ \frac{\mathrm{W}}{\mathrm{K\,m}}$
- $\mathrm{c} = 434\ \frac{\mathrm{J}}{\mathrm{K\,kg}}$
- $\mathrm{p} = 7832\ \frac{\mathrm{kg}}{\mathrm{m^3}}$
- $\alpha = 1.88 \times 10^{-5}\ \frac{\mathrm{m^2}}{\mathrm{s}}$



4 Analysis

4.1 Characteristic length

L_c = $L = 40$ mm

4.2 Biot number

Bi = $\frac{L_c h}{k} = 0.31298905$

4.3 Fourier number

Fo = $\frac{t\alpha}{L_c^2} = 5.64$

4.4 List of Biot numbers used in interpolating coefficients for one-term approximation for transient one-dimensional conduction through a plane wall

bi_list = [0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8
= [0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4,
0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0, 20.0, 30.0,
40.0, 50.0, 100.0]

4.5 List of Zeta_1 coefficients corresponding to list of Biot numbers for one-term approximation for transient one-dimensional conduction through a plane wall

zeta_1_list = [0.0998, 0.141, 0.1723, 0.1987, 0.2218, 0.2425, 0.2615, 0.2791, 0.2956, 0.3111, 0.3779, 0.4328
= [0.0998, 0.141, 0.1723, 0.1987, 0.2218, 0.2425, 0.2615, 0.2791, 0.2956, 0.3111,
0.3779, 0.4328, 0.4801, 0.5218, 0.5932, 0.6533, 0.7051, 0.7506, 0.791, 0.8274,
0.8603, 1.0769, 1.1925, 1.2646, 1.3138, 1.3496, 1.3766, 1.3978, 1.4149, 1.4289,
1.4961, 1.5202, 1.5325, 1.54, 1.5552]

4.6 List of C_1 coefficients corresponding to list of Biot numbers for one-term approximation for transient one-dimensional conduction through a plane wall

c_1_list = [1.0017, 1.0033, 1.0049, 1.0066, 1.0082, 1.0098, 1.0114, 1.013, 1.0145, 1.0161, 1.0237, 1.0311, 1
= [1.0017, 1.0033, 1.0049, 1.0066, 1.0082, 1.0098, 1.0114, 1.013, 1.0145, 1.0161,
1.0237, 1.0311, 1.0382, 1.045, 1.058, 1.0701, 1.0814, 1.0919, 1.1016, 1.1107,
1.1191, 1.1785, 1.2102, 1.2287, 1.2402, 1.2479, 1.2532, 1.257, 1.2598, 1.262,
1.2699, 1.2717, 1.2723, 1.2727, 1.2731]

4.7 Zeta_1, Coefficient for Biot number used in one-term approximation for transient one-dimensional conduction through a plane wall

ζ_1 = interp1d(bi_list, zeta_1_list)(Bi) = 0.5310741784037559

4.8 C_1, Coefficient for Biot number used in one-term approximation for transient one-dimensional conduction through a plane wall

C_1 = interp1d(bi_list, c_1_list)(Bi).item() = 1.0466885758998434

4.9 Dimensionless midplane (x* = 0) temperature after time, t

$\theta_0 = C_1 e^{-Fo\zeta_1^2} = 0.21329631$

4.10 T(x=0, t=t); Temperature of exterior pipe surface after time, t

$T_o = T_{inf} + \theta_0 \left(T_i - T_{inf}\right) = 318.21926$ K

4.11 Dimensionless surface (x* = 1) temperature after time, t

$\theta = \theta_0 \cos\left(\zeta_1\right) = 0.18391764$

4.12 T(x=L, t=t); Temperature of interior pipe surface after time, t

$T = T_{inf} + \theta \left(T_i - T_{inf}\right) = 320.27577$ K

4.13 Heat flux to the wall after time, t

$q_dot = h \left(T - T_{inf}\right) = -6437.1172 \ \frac{\mathrm{W}}{\mathrm{m}^2}$

4.14 Energy transferred to pipe per length after time, t

$Q_per_l = \pi D L_c c \rho \left(1 - \frac{\theta_0 \sin\left(\zeta_1\right)}{\zeta_1}\right) \left(T_i - T_{inf}\right) = -23817980 \ \frac{\mathrm{J}}{\mathrm{m}}$

5 Comments

- Lumped capacitance model is invalid for Bi >0.1.
- However, one-term approximation is valid for Fo >0.2.

6 References

- F. P. Incropera, Fundamentals of heat and mass transfer. 6th ed. John Wiley & Sons, 2007, Ch. 5, sec. 5.5-5.6, p.278-280.